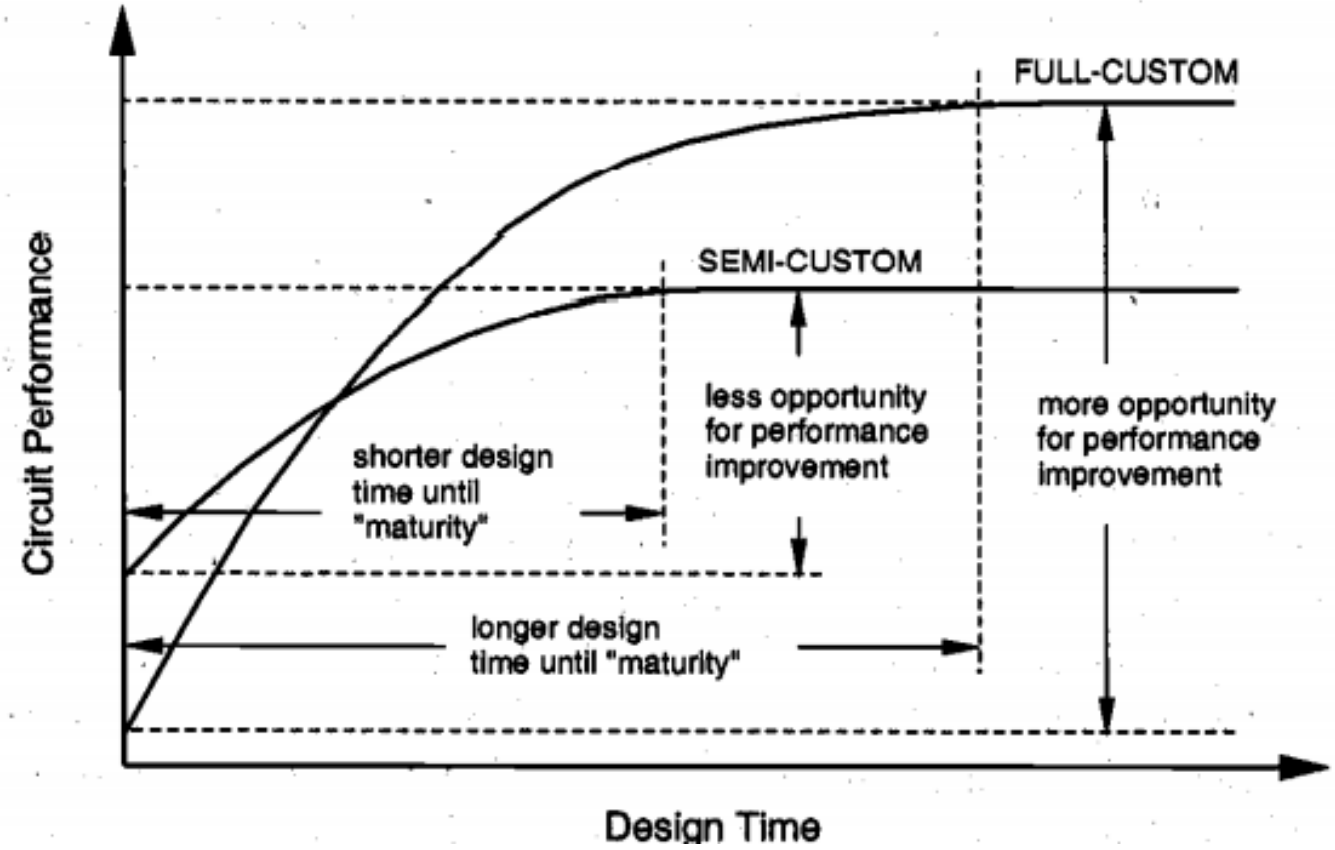
VLSI DESIGN METHODOLOGIES

- There is phenomenal growth in the structural complexity of digital integrated circuits in past few decades.
- It has been possible due to constant advances in manufacturing technology, as well as by the increasing need for integrating more complex functions on chip.
- The challenges in the design are primarily handling size of team members involved in the development of a typical VLSI product, including the development of technology, computer-aided design (CAD) tools, chip design, fabrication, packaging, testing and reliability qualification.
- The efficient organization of these efforts under a well-structured system design methodology is essential for the development of economically viable VLSI products, in a timely manner

- Further, the design complexity of logic chips increases almost exponentially with the number of transistors to be integrated.→ increased design cycle time (time period from the start of chip development until the mask-tape delivery time.)
- The majority of this design cycle time is typically devoted to achieving the desired level of chip performance at an acceptable cost, which is essential for the economical success of any competitive commercial product.
- During the course of the design cycle the circuit performance can usually be increased by design improvements; more rapidly in the beginning, then more gradually until the performance finally saturates for the particular design style and technology being used. The level of circuit performance which can be reached within a certain design time strongly depends on the efficiency of the design methodologies, as well as on the design style.

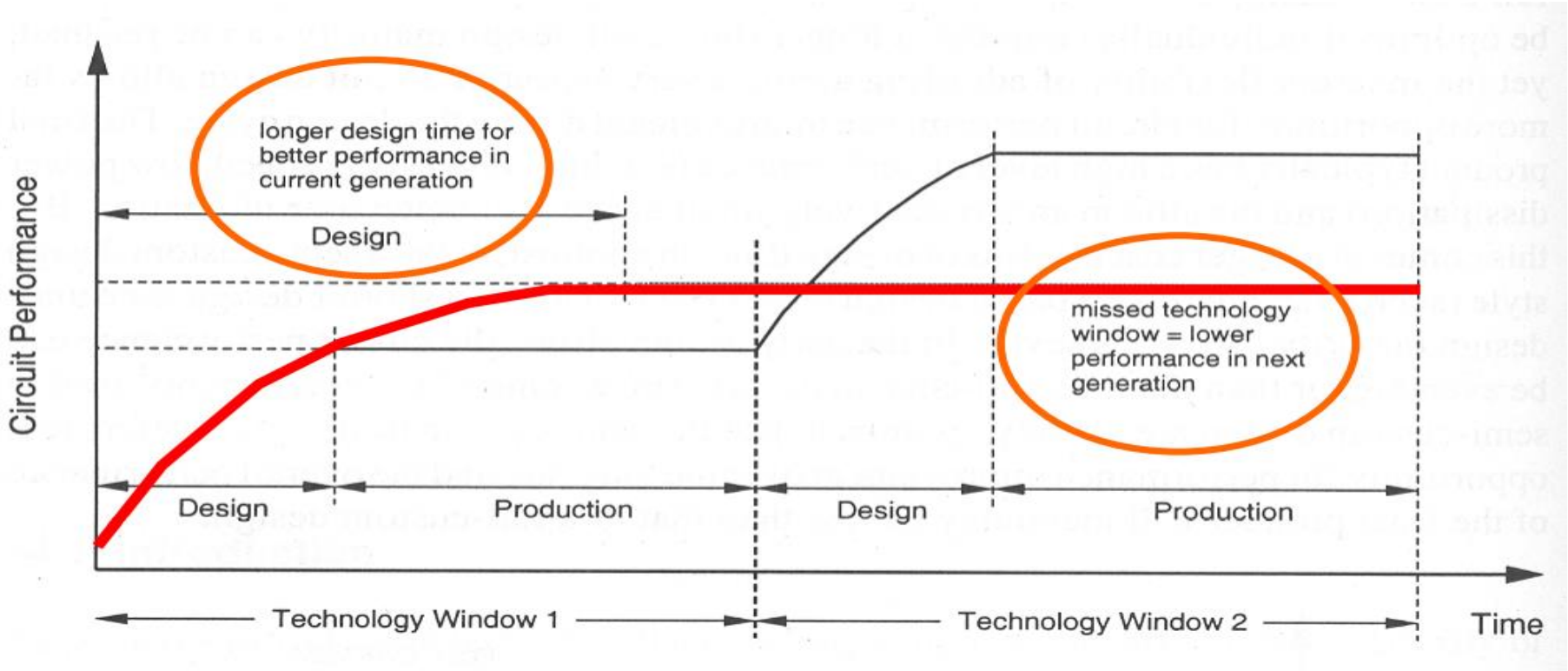
Impact of different VLSI design styles upon the design cycle time and the achievable circuit performance.

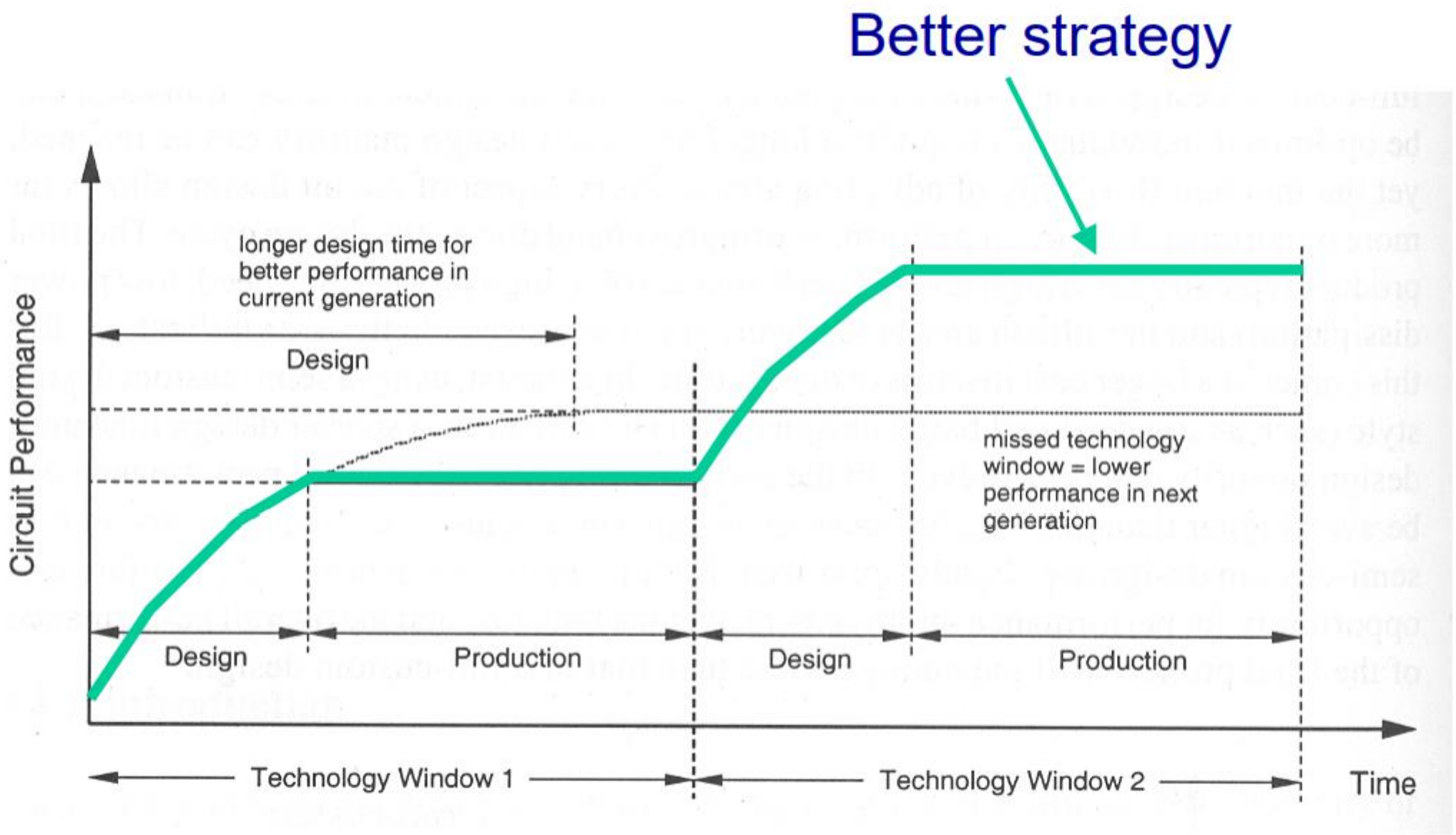


The choice of the particular design style for a VLSI product depends on the performance requirements, the technology being used, the expected lifetime of the product and the cost of the project.

- There is considerable impact of design style on circuit performance and overall cost.
- In addition to the proper choice of a VLSI design style, there are other issues which must be addressed in view of the constantly evolving nature of the VLSI manufacturing technologies. New generation of technology is introduced in approximately every two years leading to smaller device dimensions and consequently, higher integration density and higher performance.
- In order to make the best use of the current technology, the chip development time has to be short enough to allow the maturing of chip manufacturing and timely delivery of the product to customers.
- This may require that the level of logic integration and chip performance fall short of the level achievable with the current processing technology.

Progressive performance improvement of a VLSI product for each new generation of manufacturing technology. Shorter design cycle times are essential for economic viability




Better strategy
longer design time for better performance in current generation
Design
missed technology window = lower performance in next generation
Circuit Performance
Design
Production
Design
Production
Technology Window 1
Technology Window 2
Time

- It can be seen that the design cycle time of a successful VLSI product is kept shorter than what would be necessary for developing an optimum-performance chip, thus leaving enough time for the production and marketing of the chip during the current generation, or "technology window".
- When the next generation of manufacturing technology arrives, the design can be updated to take advantage of higher integration density and better performance.
- On the other hand, if the design time of a particular product is kept excessively long to achieve the highest possible performance for the current generation of technology, there is a danger of missing the next technology window.
- A longer design cycle time usually results in better overall perfornance, but this product must then remain in the market for a certain amount of time in order to recover the development costs. Thus, the advantages brought by the next generation of manufacturing technologies cannot be realized, and the product becomes less competitive.
- In reality, the design cycle of the next generation chips usually overlaps with the production cycle of the current generation chips, thereby assuring continuity. The use of sophisticated computer-aided design (CAD) tools and methodologies are also essential for reducing the design cycle time and for managing the increasing design complexity.

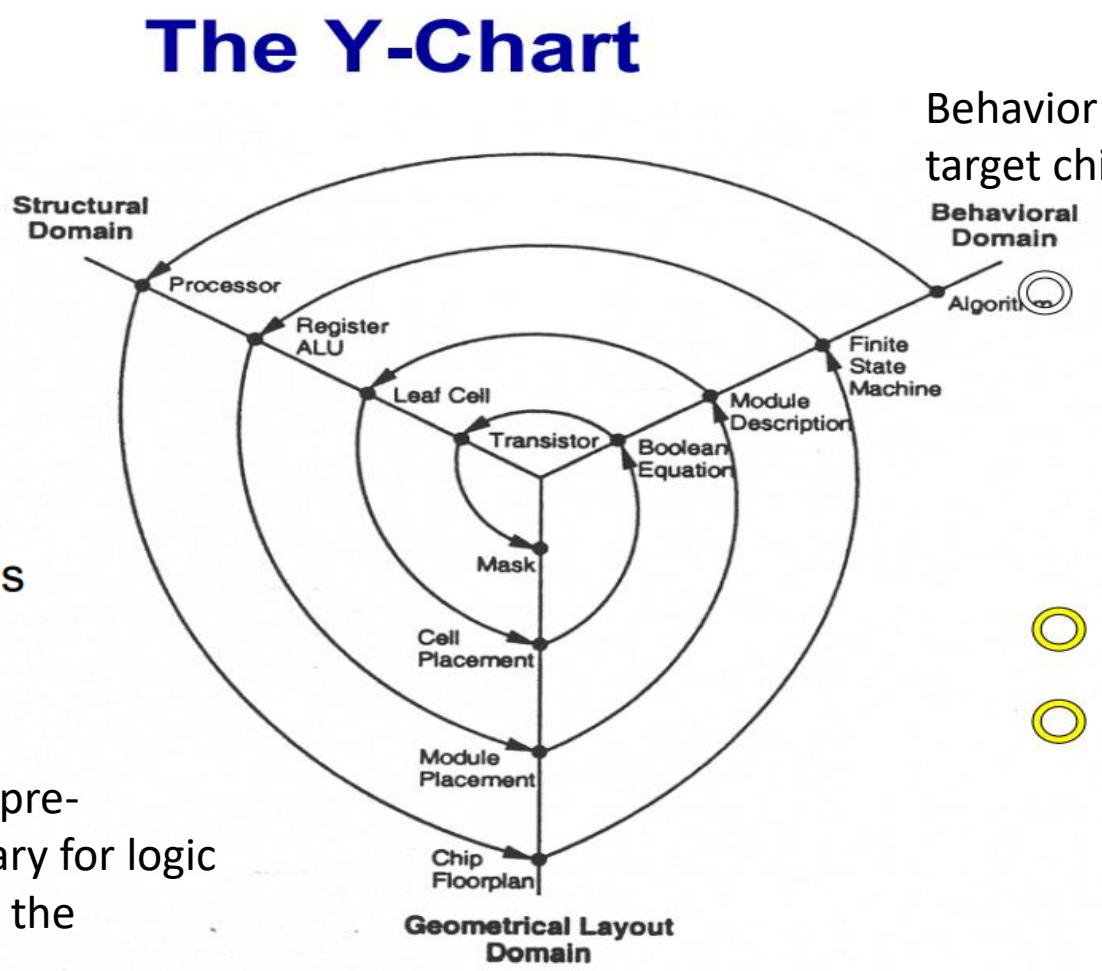
VLSI Design Flow (D. Gajski))

Registers and ALU → Functional Modules
Leaf Cell → logic gate

Cell Placement and Routing Program:
Place and interconnect the leaf Cells

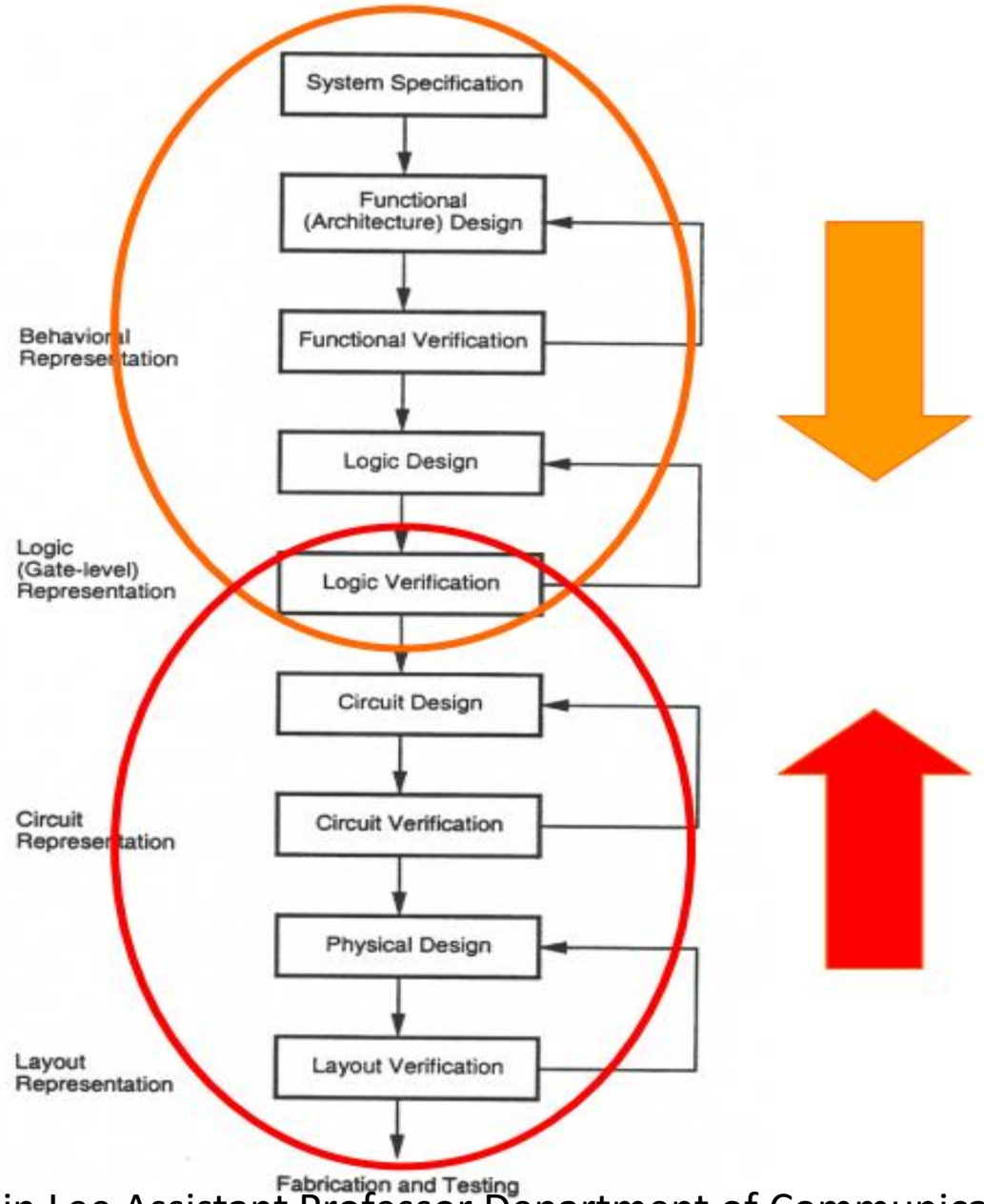
Module Placement is followed by routing Goal: Mininimze interconnect area and signal delays

Notice: There is a need for structured design methodologies to handle the high level of complexity !



In the standard cell based design style, leaf cells are pre-designed (at the transistor level) and stored in a library for logic implementation, effectively eliminating the need for the transistor level design

Simplified VLSI Design Flow



Top-down

Note that the verification of design plays a very important role in every step during this process. The failure to properly verify a design in its early phases typically causes significant and expensive re-design at a later stage, which ultimately increases the time-to-market.

Bottom-up

Top-down vs. bottom-up design

- Top-down design adds functional detail.
  - Create lower levels of abstraction from upper levels.
- Bottom-up design creates abstractions from low-level behavior.
- Good design needs both top-down and bottom-up efforts.

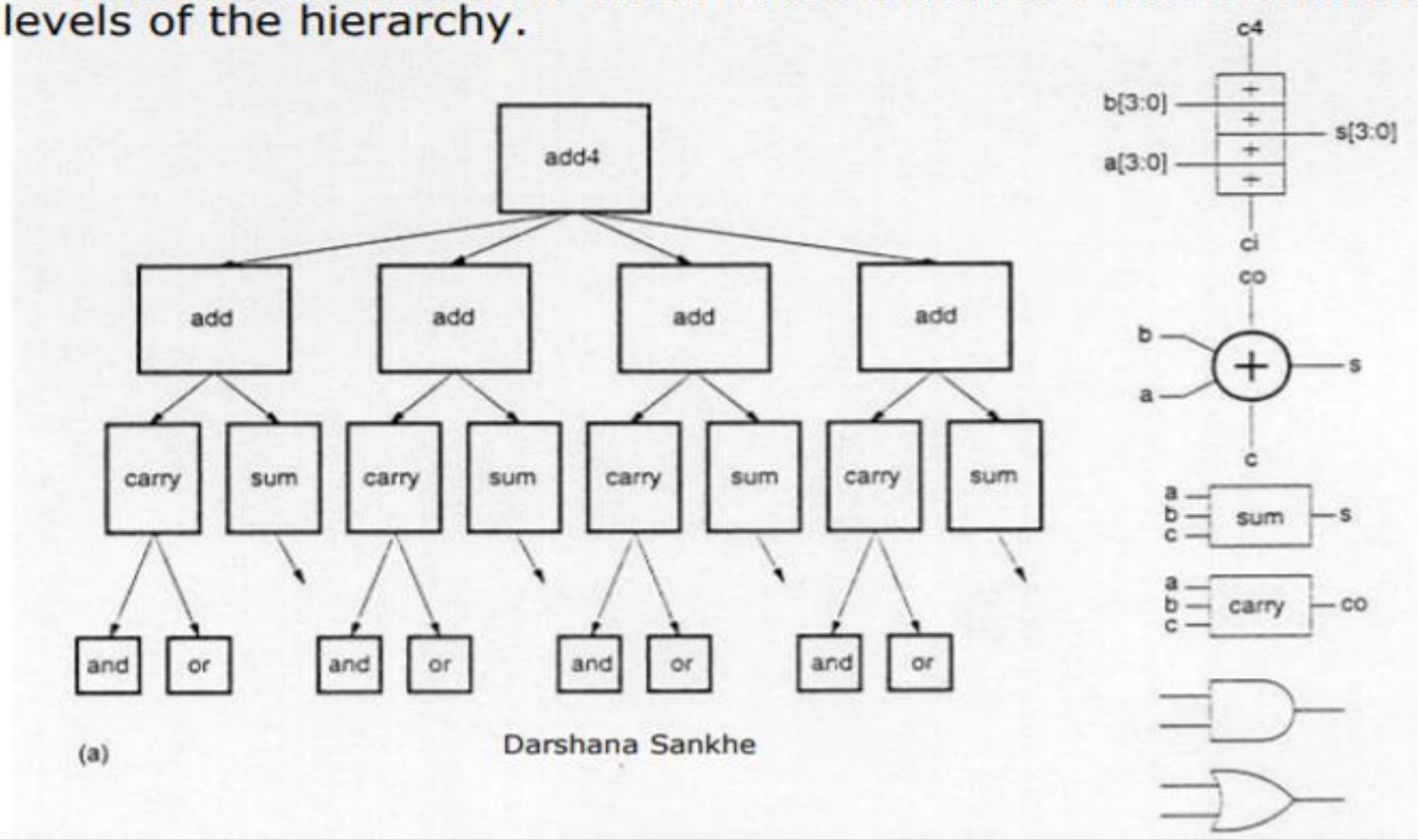
Design Hierarchy

- The use of hierarchy, or "divide and conquer" technique involves dividing a module into sub- modules and then repeating this operation on the sub-modules until the complexity of the smaller parts becomes manageable.
- This approach is very similar to the software case where large programs are split into smaller and smaller sections until simple subroutines, with well-defined functions and interfaces, can be written.
- We know that the design of a VLSI chip can be represented in three domains.
- Correspondingly, a hierarchy structure can be described in each domain separately.
- However, it is important for the simplicity of design that the hierarchies in different domains can be mapped into each other easily

Dr. Neeraj Kaushik

Design Hierarchy

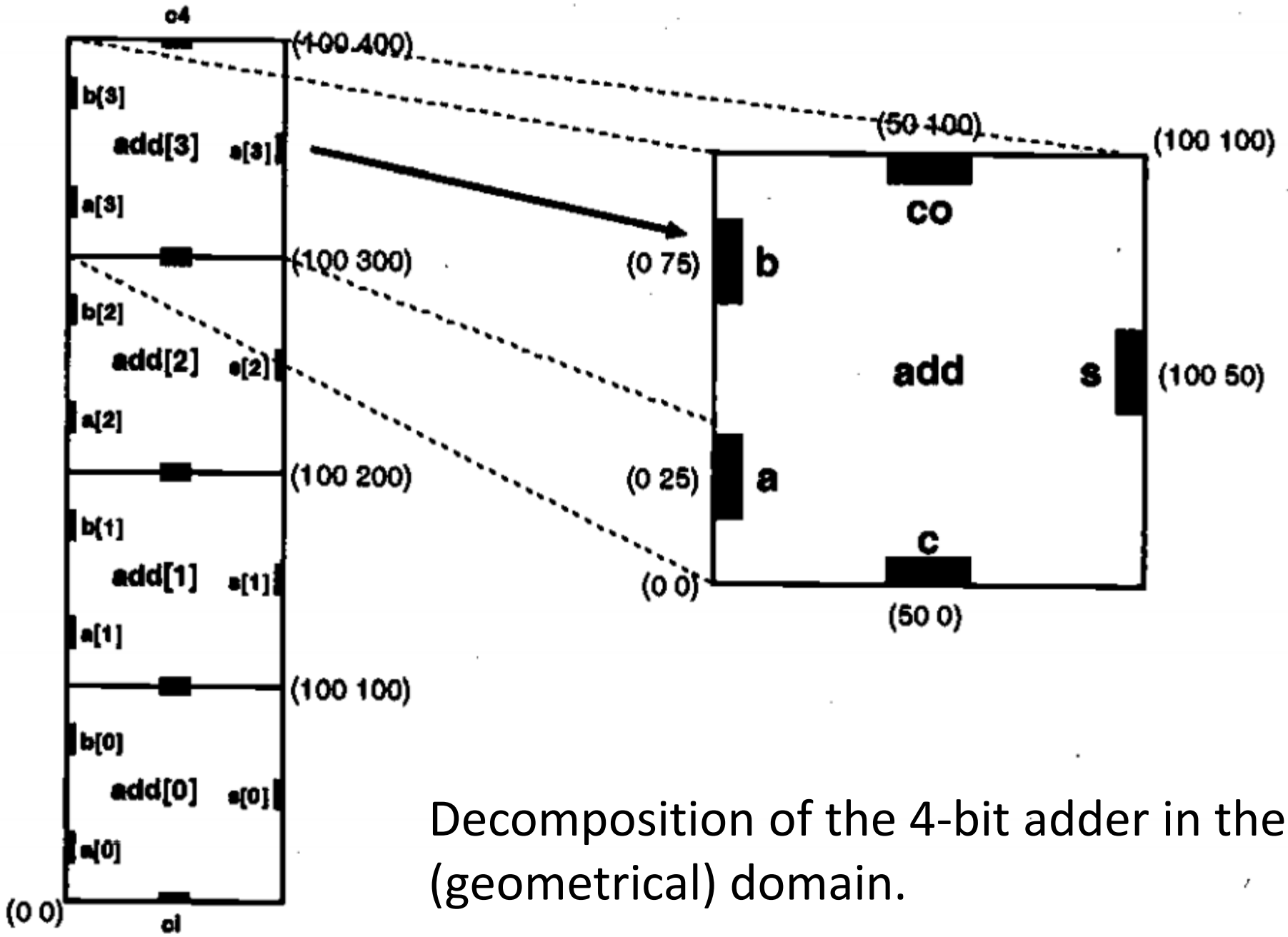
- Example: A CMOS four-bit adder
  - The adder can be decomposed progressively into one- bit adders, separate carry and sum circuits, and finally, into individual logic gates.
  - At this lower level of the hierarchy, the design of a simple circuit realizing a well-defined Boolean function is much more easier to handle than at the higher levels of the hierarchy.



Design Hierarchy

- In the physical domain, partitioning a complex system into its various functional blocks will provide a valuable guidance for the actual realization of these blocks on chip.

- Obviously, the approximate shape and size (area) of each sub-module should be estimated in order to provide a useful floorplan.



Decomposition of the 4-bit adder in the physical (geometrical) domain.

Concepts of Regularity

- The hierarchical design approach reduces the design complexity by dividing the large system into several sub-modules.
- Usually, other design concepts and design approaches are also needed to simplify the process.

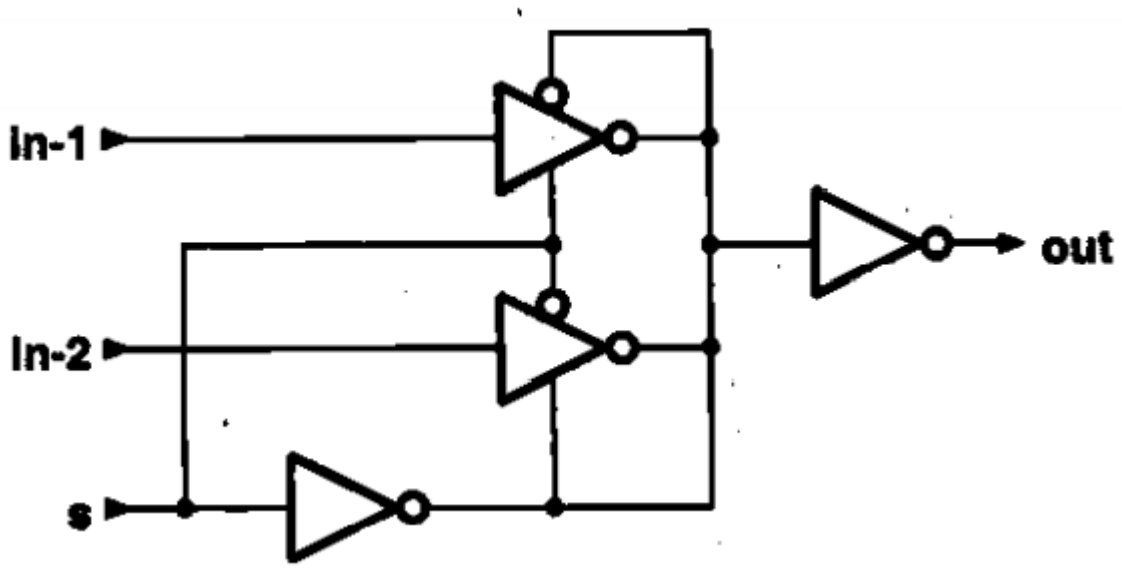
- Regularity means that the hierarchical decomposition of a large system should result in not only simple, but also similar blocks, as much as possible.

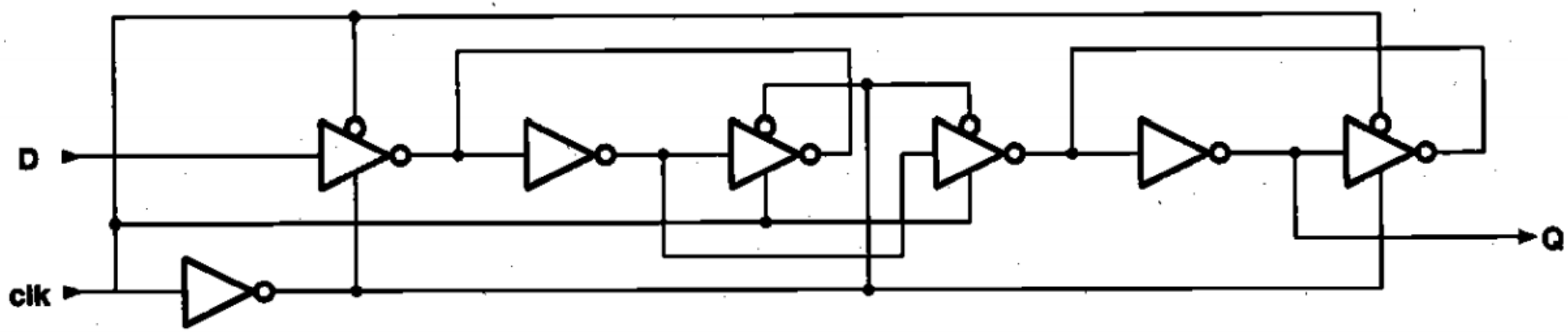
- A good example of regularity is the design of array structures consisting of identical cells - such as a parallel multiplication array, PAL, PLA, PROM.

- Regularity can exist at all levels of abstraction:
  - At the transistor level, uniformly sized transistors simplify the design.
  - At the logic level, identical gate structures can be used, etc.

- If the designer has a small library of well-defined and well-characterized basic building blocks, a number of different functions can be constructed by using this principle.

- Regularity usually reduces the number of different modules that need to be designed and verified, at all levels of abstraction.





Regular design of (a) 2-1 MUX and (b) DFF, using inverters and tri-state buffers as basic building blocks.

Concepts of Modularity

- Modularity in design means that the various functional blocks which make up the larger system must have well-defined functions and interfaces.
- Modularity allows that each block or module can be designed relatively independently from each other, since there is no ambiguity about the function and the signal interface of these blocks.
- All of the blocks can be combined with ease at the end of the design process, to form the large system.
- The concept of modularity enables the parallelization of the design process, reduces development time.
- It also allows the use of generic modules in various designs - the well-defined functionality and signal interface allow plug-and-play design.

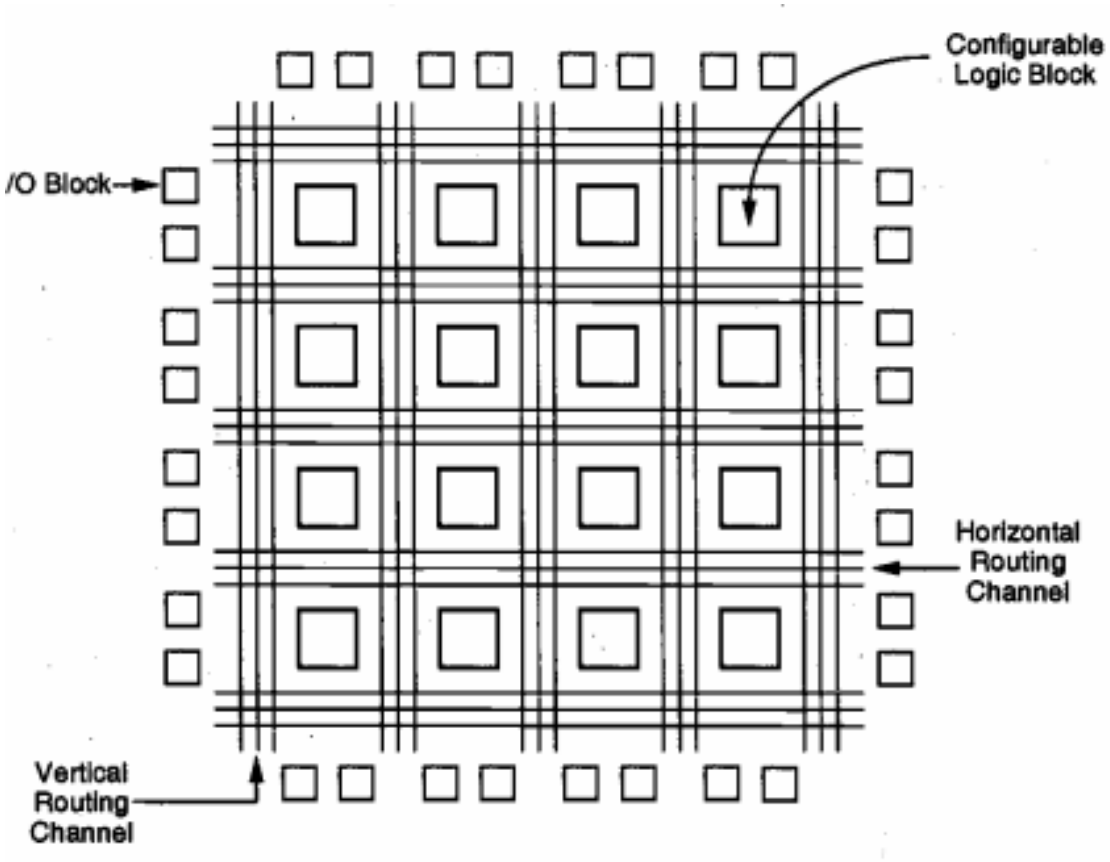
Concepts of Locality

- By defining well-characterized interfaces for each module in the system, we effectively ensure that the internals of each module become unimportant to the exterior modules.
- Internal details remain at the local level.
- The concept of locality also ensures that connections are mostly between neighboring modules, avoiding long-distance connections as much as possible.
- This last point is extremely important for avoiding excessive interconnect delays.
- Time-critical operations should be performed locally, without the need to access distant modules or signals.
- If necessary, the replication of some logic may solve this problem in large system architectures.

VLSI DESIGN STYLES:

1)Field Programmable Gate Array (FPGA):

- Fully fabricated FPGA chips containing thousands or even more, of logic gates withprogrammable interconnects, are available to users for their custom hardware programming to realize desired functionality.
- This design style provides a means for fast prototyping and also for cost-effective chip design, especially for low-volume applications.
- A typical field programmable gate array (FPGA) chip consists of I/O buffers, an array of configurable logic blocks (CLBs), and programmable interconnect structures.

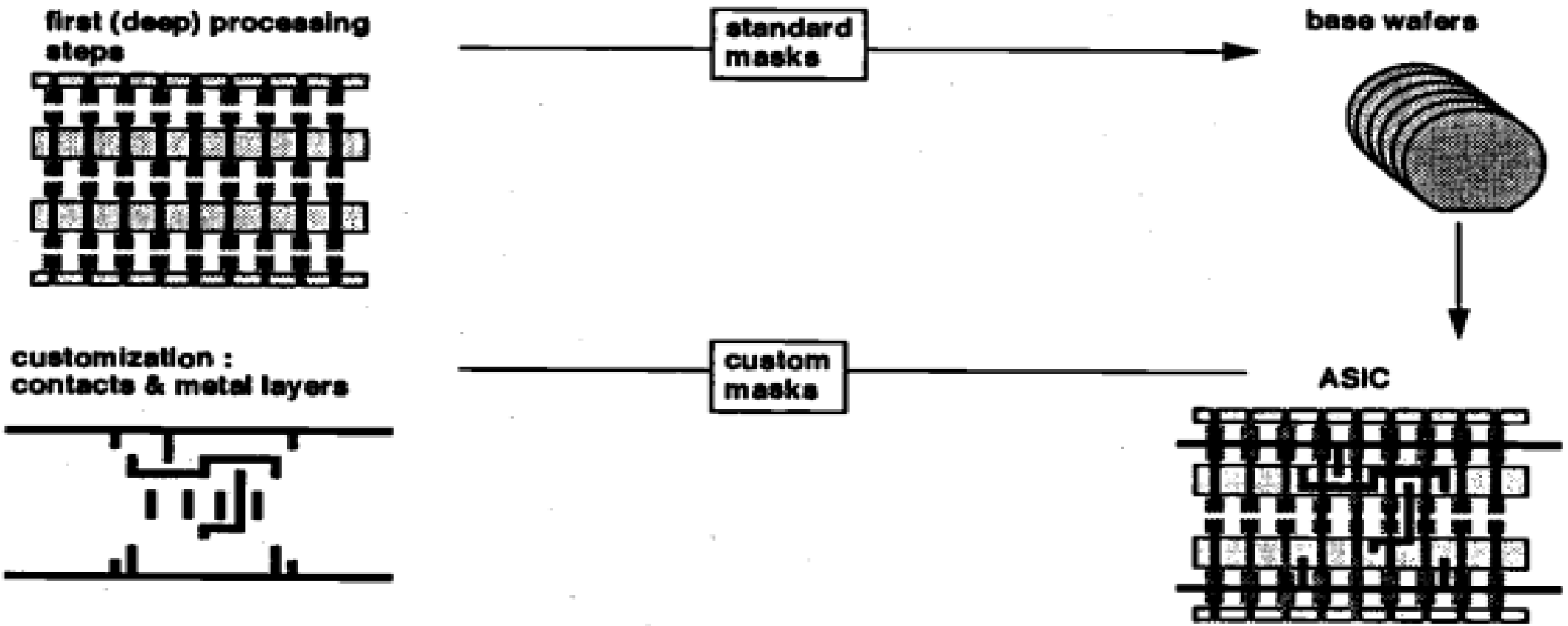


FPGA Architecture

2) Gate Array Design:

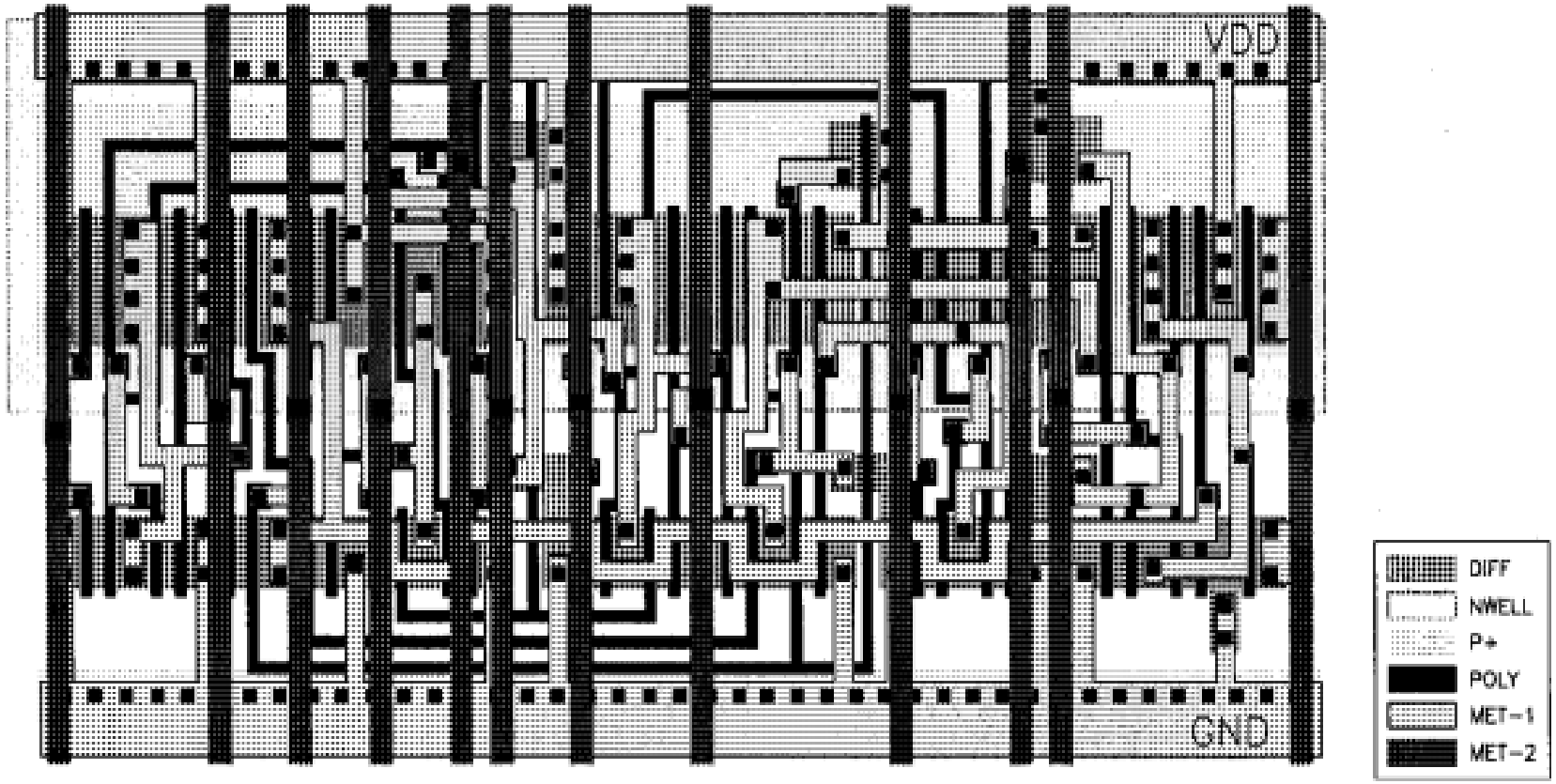
- In terms of fast prototyping capability, the gate array (GA) ranks second after the FPGA with a typical turn-around time of a few days.
- While user programming is central to the design implementation of the FPGA chip, metal mask design and processing is used for GA.
- Gate array implementation requires a two-step manufacturing process: The first phase, which is based on generic (standard) masks, results in an array of uncommitted transistors on each GA chip.
- These uncommitted chips can be stored for later customization,which is completed by defining the metal interconnects between the transistors of the array.
- Since the patterning of metallic interconnects is done at the end of the chip fabrication process, the turn-around time can still be short, a few days to a few weeks.

Basic processing steps required for gate array implementation:



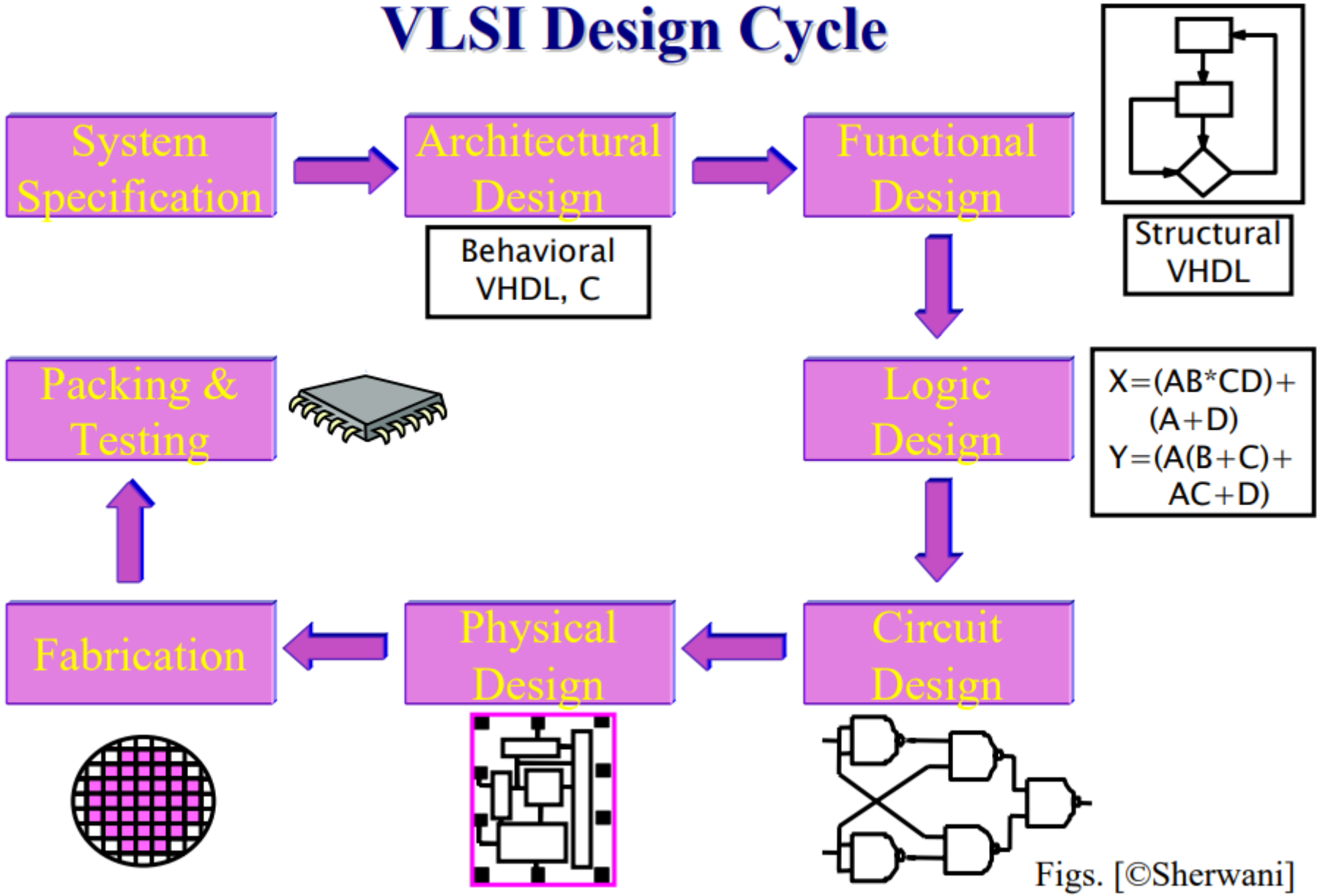
3)Standard Cell Based Design:

- The standard-cells based design is one of the most prevalent full custom design styles which require development of a full custom mask set. The standard cell is also called the polycell.
- In this design style, all of the commonly used logic cells are developed, characterized, and stored in a standard cell library.
- A typical library may contain a few hundred cells including inverters, NAND gates, NOR gates, complex AOI, OAI gates, D latches, and flip-flops.
- Each gate type can be implemented in several versions to provide adequate driving capability for different fanouts.
- For instance, the inverter gate can have standard size, double size, and quadruple size so that the chip designer can choose the proper size to achieve high circuit speed and layout density.



A standard cell layout example

VLSI Design Cycle



Figs. [©Sherwani]

VLSI Design Cycle

- **System Specification**
  - A high level representation of the system
  - Considered factors
    - » Performance
    - » Functionality
    - » Physical dimensions (die size)
  - Result Specs – size, speed, power, and functionality

Fabrication Technology

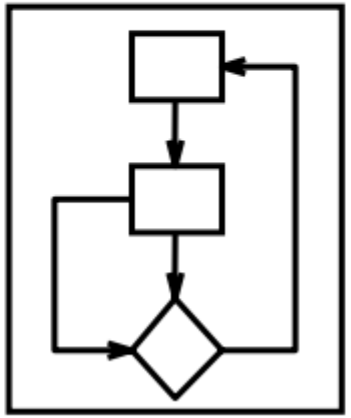
Design Techniques

VLSI Design Cycle

- **Architectural Design**
  - RISC (Reduced Instruction Set Computer) versus CISC (Complex Instruction Set Computer)
  - Number of ALUs, Floating Point Units
  - Number and structure of pipelines
  - Cache size
  - Prediction on die size, power, and speed based on existing design
  - Early estimation are very important here

VLSI Design Cycle

- **Behavioral or Functional Design**
  - Only behavior and timing without implementation issue
  - Specify behavior based on Input + output + timing
  - Fast emulation and debugging for the system



```
always @(posedge clk);
  begin
    if (enable_ == 1'b0)
      data = 0;
    else
      data = data + 1;
  end
```

VLSI Design Cycle

- **Logic Design**
  - Control flow, word widths, register allocation, arithmetic operations, and logic operations
  - RTL (Register Transfer Level) – HDL (Hardware Description Language)
    - » Verilog – most popular
    - » VHDL – Europe and Eastern
    - » Literal + Timing Information

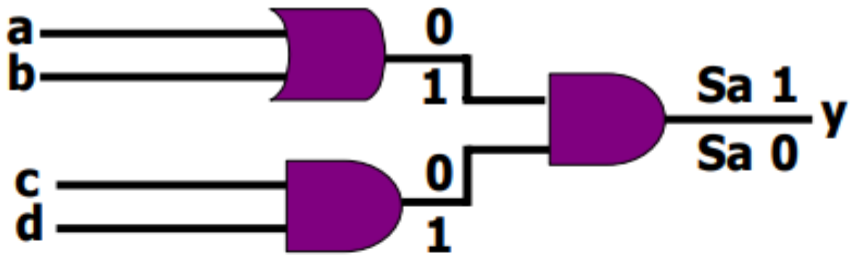
Boolean Expression

Timing Information

$$X=(AB*CD)+(A+D)$$
$$Y=(A(B+C)+AC+D)$$

VLSI Design Cycle

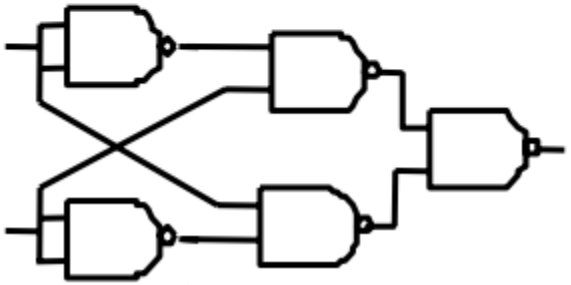
- **Logic Design**
  - More actual simulation and testing



  - High Level Synthesis: Produce a RTL description from a behavioral description of the design

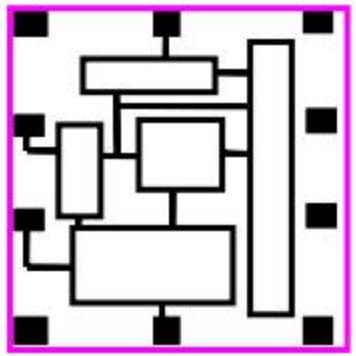
VLSI Design Cycle

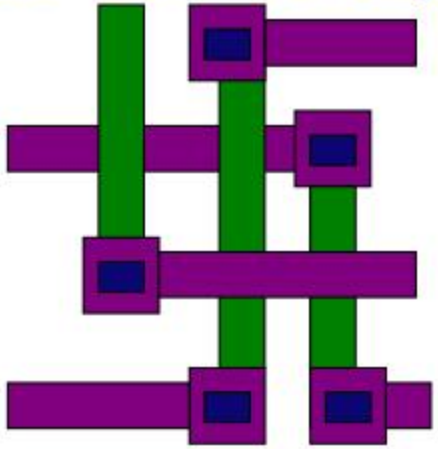
- **Circuit Design**
  - Boolean Expression → Circuit Elements (Cells, Macros, Gates, Transistors) + Interconnection
  - Each component has specific timing and power Info.
  - Circuit Simulation : Verify the correctness and timing
  - Terms – Netlist, Schematic
  - Logic Synthesis Tool s : RTL → Netlist

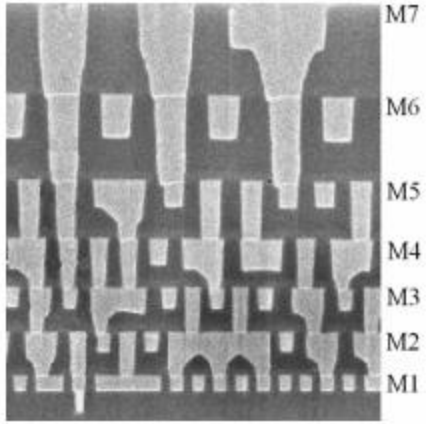


VLSI Design Cycle

- **Physical Design**
  - Netlist → Layout (Geometry Representation)
    - » Design rules of applied fabrication process
  - Layout Synthesis Tools
    - » Automatic conversion (Fully/Partially)
    - » Area and performance penalty
  - Crucial Challenges – Area/Delay

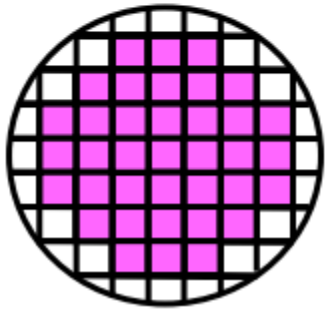






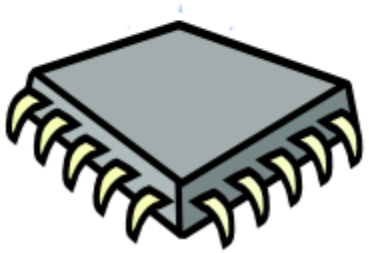
VLSI Design Cycle

- **Fabrication**
  - Layout → Photo-lithographic mask
    - » One mask for each layer
  - Wafer : Silicon crystal are grown & sliced
  - Deposition, and diffusion of various materials on the wafer : each step uses one mask
  - Term : Tape Out, 8 inch/20cm, 12 inch/30cm



VLSI Design Cycle

- **Packaging, Testing, and Debugging**
  - For PCB (Printed Circuit Board) : DIP (Dual In-line Package), PGA (Pin Grid Array), BGA (Ball Grid Array), and QFP (Quad Flat Package)
  - For MCM (Multi-Chip Modules): no packaged
  - Testing
    - » Before Package – Probe line testing
    - » After Package – Tester machine applies test patterns.



Design Quality

- Testability
  - fabricated chips should be fully testable to ensure that all the chips passing the specified chip test can be inserted into the system, either in packaged or in bare die form, without causing failures. Requirements * Generation of good test vectors * Availability of reliable test fixture at speed * Design of testable chip
- Manufacturability
  - chip yield can be calculated by dividing the number of good tested chips by the total number of tested chips.
    - Functional yield
      - Test functionality of the chip at a speed usually lower than the required chip speed.
      - The functional test weeds out the problems and Parametric yield
  - Parametric Yield
    - performed at the required speed on chips that passed the functional test. All the delay testing is performed at this stage.

Contd.

- Reliability
  - Electrostatic discharge (ESD) and electrical overstress (EOS) * Electromigration * Latch-up in CMOS I/O and internal circuits * Hot-carrier induced aging * Oxide breakdown* Single event upset * Power and ground bouncing * On-chip noise and crosstalk
- Technology Updateability
  - The design style should be chosen such that the technology update of the chip ,or functional modules for design reuse can be achieved quickly with minimal cost.
  - Designers can develop and use advanced CAD tools that can automatically generate the physical layout, the so-called silicon compilation, which meets the timing requirements with proper gate sizing or transistor sizing.

CAD Technology :

- Synthesis tools
  - VHDL
  - Verilog
- Layout tools
  - Microwind
  - Magic
  - Tanner
  - Synopsis
- Simulation and Verification
  - SPICE
  - HSPICE